World Development Sustainability 8 (2026) 100284

Contents lists available at ScienceDirect

World Development Sustainability

journal homepage: www.elsevier.com/locate/wds

Mapping poverty and food security: A spatial correlation analysis in central java

Shanty Oktavilia [a,*], Firmansyah [b], Zainur Hidayah [c], Fauzul Adzim [a], Sucihatiningsih Dian Wisika Prajanti [a], Fafurida [a]

[a] Department of Development Economics, Faculty of Economics and Business, Universitas Negeri Semarang, Indonesia
[b] Department of Development Economics, Faculty of Economics and Business, Universitas Diponegoro, Indonesia
[c] Department of Management, Faculty of Economics and Business, Universitas Terbuka, Indonesia

ARTICLE INFO

Keywords:
Poverty
Food security
Spatial analysis
Moran's I
LISA
Central java

ABSTRACT

Poverty and food security are two closely interrelated global issues and are top priorities in the Sustainable Development Goals (SDGs) agenda, particularly SDG 1 (no poverty) and SDG 2 (no hunger). This study aims to analyze the spatial correlation between poverty and food security in Central Java in the 2023–2024 period. The research method used is a quantitative descriptive approach with spatial analysis using Moran's I and Local Indicator of Spatial Association (LISA). Secondary data were obtained from the Central Statistics Agency (BPS), the National Food Agency, and administrative maps in shapefile form. The analysis was conducted using GeoDa software, by examining univariate and bivariate spatial autocorrelation patterns, as well as mapping High-High, Low-Low, High-Low, and Low-High clusters. The results show that the distribution of poverty and food security indices in Central Java is not random, but rather forms a clustered pattern. Bivariate analysis shows a negative spatial correlation, where areas with high poverty rates tend to be associated with low food security. The LISA Bivariate Map identifies clusters of High-High areas concentrated in the southern and coastal regions, while urban areas tend to be in the Low-Low category with relatively better socio-economic conditions. The implication of these findings is the importance of spatially based development policies that integrate poverty alleviation programs with improving food security. Therefore, spatially integrated policy interventions are crucial. Local governments are recommended to prioritize targeted programs in High–High areas, including improving rural food logistics and distribution infrastructure, expanding community-based microfinance and agricultural innovation programs.

1. Introduction

Poverty and food security are two closely interrelated global issues. The United Nations (UN), through its Sustainable Development Goals (SDGs), places poverty eradication (SDG 1) and food security (SDG 2) as primary goals of sustainable development. Reality shows that poor communities have limited purchasing power, access, and affordability to food, which ultimately leads to low nutritional resilience and a low quality of life. Furthermore, weak food security in a region is often associated with higher household expenditures on basic necessities, which tends to coexist with higher poverty levels. This phenomenon is not only an economic issue but also has a spatial dimension, reflecting unequal distribution between regions [1,2].

In Indonesia, the relationship between poverty and food security is a crucial issue for regional development. Although the national poverty rate continues to decline, data from the Central Statistics Agency (BPS) shows that millions of people still live below the poverty line. This situation creates vulnerability to adequate food needs, both in terms of availability, affordability, and nutritional quality. Furthermore, the distribution of poverty and food security is uneven across regions. Some areas with high poverty rates tend to face more serious food security issues, including limited production, high food prices, and the prevalence of stunting. Studies by Rosalia & Hakim [3] and Abdullah et al. [4] confirm that low food security in many Asian countries is closely linked to socioeconomic factors and multiple spatial risks, such as floods, droughts, and political volatility.

Central Java Province is one of the regions with a relatively high poverty rate compared to other provinces on the island of Java.

* Corresponding author at: FEB UNNES Sekaran Campus Gunungpati Semarang 50229 Central Java, Indonesia.
E-mail address: oktavilia@mail.unnes.ac.id (S. Oktavilia).

https://doi.org/10.1016/j.wds.2026.100284

Available online 27 February 2026

According to the Statistics Indonesia (BPS), the percentage of poor people in Central Java will still be above the national average in 2023. This condition is reflected in the Food Security Index (FSI), with several regencies/cities in Central Java exhibiting low food security, particularly in rural areas with traditional agricultural bases and limited infrastructure access. This phenomenon is interesting to study because poverty and food security in Central Java are influenced not only by individual household factors but also have a spatial dimension. Areas with high poverty rates are often adjacent to other areas that are also food insecure, creating clustering patterns. This aligns with the findings of Lone & Mayer [5] in India and Alemu et al. [6] in Ethiopia, which emphasizes that low food security often forms spatial patterns consistent with the distribution of poverty.

Conventional statistical approaches are generally only able to see linear relationships between variables without considering the spatial dimension. However, socioeconomic phenomena such as poverty and food security are spatially correlated, meaning they are interconnected between adjacent regions. A district with a high poverty rate, for example, is often adjacent to another district that is also food insecure. This creates a spatial clustering effect that cannot be comprehensively captured using conventional regression analysis alone. Therefore, a special approach is needed, namely spatial analysis using Moran's I and the Local Indicator of Spatial Association (LISA), which can measure spatial autocorrelation and map interregional linkages in greater detail. This approach aligns with research by Cai et al. [7] and Bofa & Zewotir [8], which emphasizes the importance of spatial methods in explaining the dynamics of food security across regions and time.

Spatial analysis methods allow researchers to identify spatial patterns of poverty and food security, whether they tend to be clustered, dispersed, or random. Furthermore, spatial analysis also allows for understanding the relationship between poverty levels in a region and food security conditions in surrounding areas. The LISA (Local Indicator of Spatial Association) method is appropriate for use in this research because it is able to identify and map local spatial linkage patterns between poverty levels and food security, thereby revealing clusters of regions that have similar or different socio-economic characteristics geographically. Furthermore, the analysis results can map regions into four main categories: High-High (high poverty - high food security), Low-Low (low poverty–low food security), High-Low, and Low-High. This mapping is important for generating more targeted policy recommendations, such as poverty alleviation programs integrated with increasing food access in vulnerable areas. This approach aligns with the findings of Feizizadeh et al. [9] and Hameed et al. [10], which emphasize that spatial analysis can provide more accurate food security maps that can be used as a basis for policy formulation.

The urgency of this research lies in the fact that Central Java Province is among the three provinces with the highest number of poor people in Indonesia. Second, there is significant variation in food security between regions, with districts on the north and south coasts tending to be more vulnerable than large cities or food production centers. This situation also aligns with research by Qiao et al. [11], Lv et al. [12], and Wang et al. [13], which shows that the post-crisis period (whether health, political, or climate) is a crucial moment for spatial remapping of food security.

Reviewing previous research reveals a research gap that underpins the importance of this study. Most previous studies on poverty and food security in Indonesia have used descriptive approaches or linear regression without incorporating spatial dimensions, thus failing to explain how interregional relationships influence socioeconomic conditions. The few studies that do employ spatial analysis have focused more on public health aspects such as stunting or nutritional distribution, while the direct link between poverty and the food security index at the district/city level in Central Java has not been widely explored [14–16]. Investigating the spatial relationship between poverty and food security at the district level is essential because both phenomena are inherently place-based and shaped by interregional interdependence rather than isolated local conditions. In Central Java, poverty and food security do not distribute evenly across space; instead, they exhibit clear geographic disparities driven by differences in agroecological conditions, infrastructure accessibility, market integration, and regional economic structures. Districts experiencing high poverty are often geographically proximate to areas facing food insecurity, suggesting the presence of spatial spillover effects where socioeconomic vulnerabilities transcend administrative boundaries. Conventional non-spatial analyses are unable to capture these inter-district linkages and therefore risk oversimplifying the poverty–food security nexus. Conducting spatial analysis at the district level is particularly important in the context of Indonesia's decentralized governance system, where development policies and food interventions are implemented locally but their impacts frequently extend across neighboring regions. This research is essential to identify spatial clusters of vulnerability and resilience, enabling policymakers to distinguish between structurally food-insecure regions and those affected by surrounding conditions. The contribution of this study lies in providing district-level empirical evidence on the spatial correlation between poverty and food security using Moran's I and LISA methods, offering a more nuanced understanding of territorial inequality in Central Java. Therefore, this study aims to fill this gap by specifically examining the spatial correlation between poverty and food security in Central Java in 2023–2024, thus providing a more comprehensive and relevant understanding to support regional development policies. The specific objective of this study is to analyze the spatial relationship between poverty levels and food security in Central Java Province using a quantitative approach based on spatial analysis.

2. Food security and poverty

This section provides the conceptual and analytical foundation for examining the relationship between poverty and food security. It first outlines the theoretical and operational definitions of food security, including its multidimensional pillars and measurement through the Food Security Index (FSI). The section then discusses poverty as a multidimensional socioeconomic phenomenon, emphasizing its spatial characteristics and regional disparities. By reviewing key theoretical perspectives and empirical findings, this section establishes the conceptual linkage between poverty and food security and highlights why a spatial approach is necessary to understand their interaction at the district level in Central Java.

2.1. Food security

The concept of food security is one of the most crucial global issues in the context of sustainable development and social well-being. According to the definition put forward by the FAO (1996) at the World Food Summit, food security exists when all individuals, at all times, have physical, social, and economic access to sufficient, safe, and nutritious food to meet the needs of an active and healthy life. This concept emphasizes four key pillars of food security: availability, access, utilization, and stability. Availability reflects the adequate quantity and distribution of food; access relates to an individual's economic and physical ability to obtain food; utilization refers to how food is consumed and biologically utilized; and stability emphasizes the continuity of these four aspects over time.

In the Indonesian context, food security measurement has been institutionalized through the Food Security Index (FSI), developed by the National Food Agency and the Central Statistics Agency (BPS). The FSI integrates social, economic, and environmental indicators such as poverty levels, normative consumption versus clean production, access to clean water, and women's education levels. Through this approach, food security is understood not only as a result of the agricultural sector but also as a reflection of the overall socio-economic conditions of society.

Theoretically, food security is closely linked to human development theory through the concept of the entitlement approach, where hunger is not caused solely by aggregate food shortages, but also by limited property rights and access to resources that enable individuals to obtain food. Therefore, aspects of social inequality, poverty, and resource distribution are key determinants in food security analysis.

Several previous studies have shown that food security is often spatially related to a region's socio-economic conditions. This spatial approach can uncover regional variations undetected by conventional analysis. For example, areas with high poverty rates tend to exhibit low food security, forming a high-high cluster spatial pattern [17,18]. However, in some contexts, such as Central Java, this pattern can be more heterogeneous, influenced by geographic factors such as agro-ecological conditions, infrastructure, and levels of urbanization. In the context of this research, food security is positioned as an outcome indicator that is closely associated with various socio-economic conditions, including poverty levels. Spatial analysis, specifically using Moran's I and the Local Indicator of Spatial Association (LISA) methods, was used to map the spatial distribution of food security and identify clusters of areas with food insecurity. This approach allows for a more comprehensive understanding of the spatial relationship between poverty and food security at the regional level, such as Central Java Province.

2.2. Poverty

Poverty is a multidimensional phenomenon that reflects not only limited income but also limited access to resources, education, health, and social participation. According to the World Bank (2018), poverty is defined as a condition in which an individual is unable to meet minimum basic needs such as food, clothing, shelter, education, and health. Meanwhile, the UNDP (2019) emphasizes the dimension of poverty in the context of human deprivation, namely limitations in human capabilities to lead a dignified and productive life.

In the context of development economic theory, poverty is often explained through two main approaches: absolute poverty and relative poverty. The absolute approach measures poverty based on the threshold of basic needs, while relative poverty considers economic disparities between individuals in society. The concept of the deprivation trap explains that poverty is not solely caused by economic factors, but also by institutional weaknesses, geographic isolation, and social vulnerability.

From a spatial perspective, poverty is unevenly distributed across regions. Geographic factors such as topography, access to infrastructure, and the condition of natural resources also determine a community's ability to develop productive economic activities. The New Economic Geography theory explains that the accumulation of economic activity in certain areas creates spatial inequality, which widens economic disparities between regions. In the Indonesian context, coastal and urban areas tend to have greater economic opportunities than inland or rural areas, which often rely on subsistence agriculture.

Several empirical studies also demonstrate a spatial link between poverty and food security. Areas with high poverty rates tend to be spatially associated with lower household food security, particularly in regions with limited access to productive land and markets. Similar findings also confirm that poverty is spatially correlated, meaning that poor areas tend to be close to one another due to the effects of interregional dependence [19,20].

In the context of Central Java Province, poverty remains a significant issue affecting community food security. According to data from the Central Statistics Agency (2024), the percentage of poor people in Central Java reached 10.8 percent, with the highest concentration in inland and southern regions such as Wonosobo, Banjarnegara, and Kebumen. Conversely, urban areas such as Semarang and Surakarta have lower poverty rates, reflecting significant spatial inequality.

Spatial analysis in this study is crucial for mapping patterns and interregional linkages based on poverty levels and food security. The use of a bivariate spatial correlation approach through Moran's I and LISA cluster maps will help identify regions with significant spatial correlations, such as areas with a High-High pattern (high poverty and high food security) or Low-Low (low poverty and low food security). The results of this analysis not only provide an empirical contribution to understanding the spatial relationship between poverty and food security at the regional level but also have policy implications for regional development based on welfare and food self-sufficiency.

Despite the extensive literature examining poverty and food security, several research gaps remain apparent. First, most empirical studies in Indonesia rely on non-spatial or purely descriptive approaches, which ignore spatial interdependencies between regions and fail to capture how poverty and food insecurity in one district may impact neighboring regions. Second, existing spatial studies largely focus on health outcomes such as stunting or agricultural production, rather than explicitly analyzing the spatial relationship between poverty and the composite food security index at the district level. Furthermore, limited attention has been paid to Central Java, despite its striking regional disparities and persistent pockets of poverty. To address these gaps, this study applies a district-level spatial analytical framework using Moran's I and Local Indicators of Spatial Association (LISA) to examine global and local spatial correlations between poverty and food security. This study not only identifies spatial clusters and spillover effects but also provides policy-relevant insights for designing spatially targeted and place-based interventions in the context of decentralized regional development.

3. Methods

3.1. Data and data sources

This study uses a quantitative descriptive method to examine spatial correlation and analyze the spatial relationship between poverty and food security variables in adjacent areas in Central Java Province. The study population covers all regencies/cities in Central Java with the main indicators being the percentage of poor people and the food security index (FSI) in 2023–2024. Data from previous years lacks consistent spatial format and directly comparable measurement indicators, so the use of the most recent two years is considered the most valid to maintain methodological consistency. The authors acknowledge that the relatively short time span may limit this study's ability to capture temporal dynamics or long-term trends. However, this study focuses on providing a current spatial portrait of the relationship between poverty and food security in Central Java, thus using the most recent and up-to-date data.

3.2. Research variables

Food security is treated as a dependent variable in this analysis because it is conceptually considered an outcome influenced by various socioeconomic factors, one of which is poverty level [21,22]. Within the theoretical framework of this study, poverty plays a key role in influencing the dimensions of food access and utilization, making it relevant for analysis as a factor potentially explaining spatial variations in food security across regions. The Food Security Index (FSI), developed by the National Food Agency, is a comprehensive measurement tool for assessing a region's food security status based on three main dimensions: food availability, access, and utilization. The index is compiled using nine key indicators representing various social, economic, and health aspects that influence food security. (Table 1).

The poverty percentage variable in this study is used as the main indicator to describe the level of community welfare in each district/city in Central Java Province. The use of the poverty percentage as the main indicator of community welfare in this study is based on both conceptual and practical considerations. Conceptually, poverty reflects the most fundamental dimension of welfare, namely the ability of households to meet basic needs, particularly food consumption, which is directly

Table 1
Food Security Index Indicators.

No	Indicator	Targeted Aspects of Food Security
1	The ratio of normative consumption per capita to net production	Food availability
2	Present population living in vulnerable economic conditions	Food access/affordability
3	Percentage of households with a proportion of expenditure on food of >65 % of total expenditure	Food access/affordability
4	Percentage of households without access to electricity	Physical access/ infrastructure & food utilization
5	The average length of schooling for girls is over 15 years	Food utilization (social capacity)
6	Percentage of households without access to clean water	Food utilization (quality & sanitation)
7	The ratio of the number of residents per health worker to the population density level	Food utilization (public health)
8	Percentage of toddlers with below standard height (stunting)	Utilization of food (nutrition)
9	Life expectancy at birth	Food utilization (general welfare)

Source: National Food Agency, 2025.

linked to food access and utilization [23]. In the Indonesian context, the poverty rate calculated by Central Statistics Agency (BPS), is derived from household expenditure relative to a poverty line that is dominated by food-related components, making it a relevant proxy for welfare conditions closely associated with food security. Practically, the poverty percentage offers a standardized, consistently measured, and policy-relevant indicator that is comparable across districts and over time, which is essential for spatial analysis. Unlike composite welfare indices that may obscure localized deprivation due to aggregation, the poverty rate captures absolute deprivation and allows clearer identification of spatial vulnerability and spillover effects between neighboring districts. Therefore, using poverty percentage as the primary welfare indicator enables this study to more accurately examine spatial disparities and interregional linkages in the poverty–food security relationship at the district level in Central Java.

Data are taken from the official publication of the Central Statistics Agency (BPS), specifically the Central Java Province Poverty Data and Information for 2023–2024. The poverty percentage is calculated as the proportion of the population with monthly per capita expenditure below the poverty line compared to the total population.

The sampling technique used is non-probability sampling with a saturated sampling approach, so that all members of the population (35 regencies/cities in Central Java) are used as research samples. Secondary data were obtained from various official agencies, namely the Central Statistics Agency (BPS) of Central Java Province, the National Food Agency, as well as documents from the Ministry of Agriculture and the Ministry of Health of the Republic of Indonesia through official government websites. In addition, spatial data in the form of administrative maps, topographic maps, and SHP files were downloaded from official portals such as Inageoportal.

3.3. Data analysis methods

Data processing was performed using GeoDa software to determine spatial patterns and spatial correlations between variables. Poverty rate and food security index data were integrated into a district/city administrative boundary shapefile, then a spatial weighting matrix was created using the queen contiguity method, which considers both edge and corner connectivity between regions.

The analysis was conducted using the Moran's Index (Univariate and Bivariate) and Local Indicator of Spatial Association (LISA) approaches [24,25].

1) The Univariate Moran's I test is used to determine the spatial pattern of each variable (poverty and food security), whether it is clustered, dispersed, or random.

$$I = \frac{\mathrm{n}}{\mathrm{W}} \cdot \sum\nolimits_{i=1}^{\mathrm{n}} \sum\nolimits_{j=1}^{\mathrm{n}} \mathrm{wij} \, \frac{(\mathrm{xi}-\bar{\mathrm{x}})(\mathrm{xj}-\bar{\mathrm{x}})}{\sum_{i=1}^{\mathrm{n}} \ (\mathrm{xi}-\bar{\mathrm{x}})^2}$$

Information:
Information:

a. n= number of observation units (districts/cities in Central Java)
b. xi= variable value in region i (e.g. percentage of poverty)
c. $\bar{\mathrm{x}}$= average of variable
d. $\mathrm{w_{ij}}$= spatial weight between regions i and j (queen contiguity)
e. $W = \sum_{i=1}^{\mathrm{n}} \ \sum_{j=1}^{\mathrm{n}} \mathrm{w_{ij}}$ total spatial weight

Interpretation of Moran's I value:

a. $I > 0$ = clustered pattern
b. $I < 0$ = dispersed pattern
c. $I \approx 0$ = random pattern

2) The Moran's I Bivariate Test is used to analyze the spatial relationship between poverty levels and food security conditions between regions, with the results being positive or negative correlation values.

$$\mathrm{Ixy} = \frac{\mathrm{n}}{\mathrm{W}} \cdot \sum\nolimits_{i=1}^{\mathrm{n}} \sum\nolimits_{j=1}^{\mathrm{n}} \mathrm{w_{ij}} \, \frac{(\mathrm{xi}-\bar{\mathrm{x}})(\mathrm{yj}-\bar{\mathrm{y}})}{\sqrt{\sum_{i=1}^{\mathrm{n}} \ (\mathrm{xi}-\bar{\mathrm{x}})^2 \cdot \sum_{j=1}^{\mathrm{n}} \ (\mathrm{yj}-\bar{\mathrm{y}})^2}}$$

Information:

a. xi = value of poverty variable in region i
b. yj= value of food security variable in region j
c. $\bar{\mathrm{x}}\bar{\mathrm{y}}$ = average of each variable
d. $\mathrm{w_{ij}}$, W = spatial weight and total weight as in univariate

Interpretation:

a. Ixy > 0= high poverty is associated with high food security (or vice versa) in adjacent areas (positive relationship).
b. Ixy < 0 = there is no spatial correspondence between poverty and food security (negative relationship).

3) Local Indicator of Spatial Association (LISA) is used to identify local spatial clusters (Hotspots/Coldspots) in each region.

$$I_i = \frac{(xi-\bar{x})}{m_2} \sum\nolimits_{j=1}^{n} \ w_{ij}(xj-\bar{x})$$

With

$$m_2 = \frac{\sum_{i=1}^{n} \ w_{ij}(xi-\bar{x})^2}{n}$$

Information:

a. $\mathrm{I_i}$ = LISA value for region iii
b. $\mathrm{m_2}$= average variance
c. The LISA interpretation is shown in the cluster map:

1. High-High (HH) = high value areas surrounded by high value areas
2. Low-Low (LL) = low value areas surrounded by low value areas
3. High-Low (HL) = areas with high values surrounded by areas with low values
4. Low-High (LH) = areas with low values surrounded by areas with high values

This study uses the first-order Queen contiguity method because this approach is considered most appropriate for representing direct spatial relationships between regions that share administrative boundaries, both edges and points. The selection of the first order is intended to capture the strongest and most geographically relevant spatial interactions, namely the influence between regions that are physically adjacent. The use of a higher contiguity order has the potential to over-expand spatial relationships and obscure local effects that are the main focus of the analysis, so the first order is considered most appropriate for the research objectives that emphasize the identification of spatial clusters at the district/city level in Central Java. The results of the analysis are presented in the form of Moran's Scatterplot which is divided into four quadrants (High-High, Low-High, Low-Low, High-Low) as well as spatial maps in the form of cluster maps and significance maps from Bivariate LISA (Fig. 1).

To ensure that the spatial results are not driven by a specific choice of spatial weights, this study conducts a robustness (sensitivity) analysis on the effectiveness of using a first-order Queen contiguity matrix. While Queen(1) is theoretically appropriate for district-level administrative units because it captures direct interaction through shared borders and vertices, spatial dependence can be sensitive to alternative definitions of neighbourhoods. Therefore, we re-estimate the univariate and bivariate Moran's I and the corresponding LISA cluster maps under several alternative spatial weighting schemes: (i) Rook contiguity (order 1), which restricts neighbours to shared borders only; (ii) Queen contiguity (order 2), to test whether extending neighbourhoods changes the strength and direction of spatial association; and (iii) distance-based weights / k-nearest neighbours (e.g., $k = 4$ or 5) to account for potential functional linkages that may not be fully represented by administrative contiguity. The robustness assessment is evaluated based on (a) the stability of Moran's I sign and statistical significance across weight matrices, (b) the consistency of key LISA clusters (HH/LL/HL/LH) in terms of both location and category, and (c) the proportion of districts whose cluster classification changes when the weights are altered. Consistent results across these specifications indicate that the observed spatial association between poverty and food security in Central Java is robust and not an artefact of a single spatial weights definition.

Robustness checks using alternative spatial weight matrices yield consistent results; therefore, the discussion focuses on estimates based on the first-order Queen contiguity matrix. Table 2

The robustness check confirms that the main findings are not sensitive to the choice of spatial weight matrix. Across alternative specifications rook contiguity, higher-order queen contiguity, and distance-based k-nearest neighbours the sign and general significance pattern of Moran's I remain stable for both poverty and food security variables. Although the strength of spatial autocorrelation slightly varies across matrices, particularly in 2024, the overall conclusions regarding the presence (or weakening) of spatial clustering remain unchanged. This consistency indicates that the use of a first-order Queen contiguity matrix provides a reliable and theoretically appropriate representation of spatial dependence at the district level in Central Java.

4. Results and discussion

4.1. Overview of poverty distribution and food security in central java province

Figs. 2 and 3 below compare the percentage of the poverty and the Food Security Index (FSI) in each district/city in Central Java Province for 2023 and 2024. Data were obtained from the Central Statistics Agency (BPS) and the National Food Agency, which present spatial indicators related to welfare and food security. This visualization aims to illustrate the relationship between poverty distribution patterns and food security levels across regions and to track changes that have occurred over a two-year period.

In 2023, most regencies/cities in Central Java showed relatively high Food Security Index levels, ranging from 80 to 95 percent, indicating adequate food availability and access in most areas. However, there was significant variation between regions, with some areas, such as Wonosobo, Brebes, and Pemalang, showing lower index values than others. The percentage of the poverty this year varied relatively, with poverty remaining concentrated in the southern and western regions of the province. Overall, an inverse relationship was observed, with areas with higher poverty rates tending to have lower food security indexes, indicating a spatial association between economic conditions and community food access.

In 2024, a moderate increase in the Food Security Index was observed in most districts/cities compared to the previous year. Index scores tended to be more evenly distributed, with notable improvements in areas such as Banjarnegara, Kebumen, and Blora, which were previously in the moderate category. Meanwhile, poverty rates showed a slight decline in almost all regions, although some areas, such as Brebes and Wonosobo, still had the highest poverty rates. This distribution pattern indicates a growing negative spatial correlation, whereby poverty reduction coincided with improvements in food security. This change reflects the initial success of various food intervention policies and poverty alleviation at the local level, although regional disparities remain a concern in future development planning.

Fig. 4 depicts the spatial distribution of poverty rates in Central Java Province during the 2023–2024 period. This map demonstrates clear spatial heterogeneity between regions, with geographic patterns indicating that poverty tends to be concentrated in the southern and interior regions, while food security is relatively better in the northern and central regions of the province.

In 2023, the poverty distribution map (top left) shows that areas with high poverty rates (above 12 %) are concentrated in regions such as Banjarnegara, Wonosobo, Kebumen, and Purbalingga, as well as in some mountainous areas in southern Central Java. These regions are characterized by steep topography, limited economic infrastructure, and a high dependence on traditional agriculture, which is vulnerable to climate fluctuations. In contrast, urban and peri-urban areas such as Semarang,

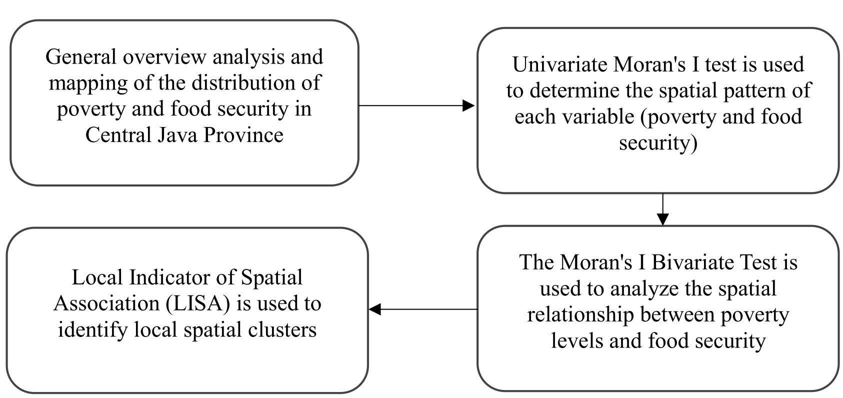

Fig. 1. Flowchart of Data Analysis in Research.

Table 2
Robustness of Moran's I Results across Alternative Spatial Weight Specifications.

Variable	Year	Spatial Weight Matrix	Moran's I (Sign)	Statistical Significance	Consistency with Queen (1)
Poverty Percentage	2023	Queen contiguity (order 1)	Positive (0.288)	Significant ($p < 0.01$)	Baseline
	2023	Rook contiguity (order 1)	Positive	Significant	Consistent
	2023	Queen contiguity (order 2)	Positive	Significant	Consistent
	2023	k-nearest neighbours ($k = 4$)	Positive	Significant	Consistent
Poverty Percentage	2024	Queen contiguity (order 1)	Positive (0.184)	Significant ($p < 0.05$)	Baseline
	2024	Rook contiguity (order 1)	Positive	Marginally significant	Consistent
	2024	Queen contiguity (order 2)	Positive	Marginally significant	Consistent
	2024	k-nearest neighbours ($k = 4$)	Positive	Marginally significant	Consistent
Food Security Index	2023	Queen contiguity (order 1)	Positive (0.164)	Significant ($p < 0.05$)	Baseline
	2023	Rook contiguity (order 1)	Positive	Significant	Consistent
	2023	Queen contiguity (order 2)	Positive	Significant	Consistent
	2023	k-nearest neighbours ($k = 4$)	Positive	Significant	Consistent
Food Security Index	2024	Queen contiguity (order 1)	Positive (0.113)	Not significant	Baseline
	2024	Rook contiguity (order 1)	Positive	Not significant	Consistent
	2024	Queen contiguity (order 2)	Positive	Not significant	Consistent
	2024	k-nearest neighbours ($k = 4$)	Positive	Not significant	Consistent

Source: Spatial Analysis Output, 2025.

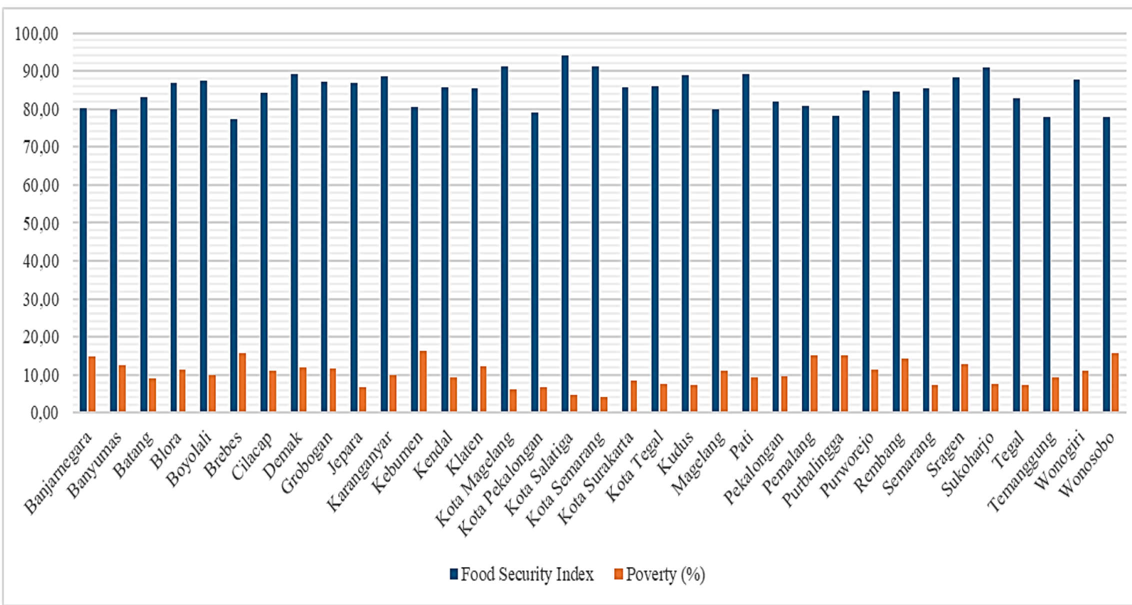

Fig. 2. Percentage of poverty and food security index of central java province in 2023.
source: central statistics agency and national food agency, 2024.

Surakarta, and Kudus show lower poverty rates. This reflects the influence of a more diverse economic structure, the availability of employment opportunities in the industrial and service sectors, and better access to markets and public facilities.

In 2024 (top right), poverty patterns show a spatial shift, increasingly forming clusters in the northern and central regions, particularly in Pati, Rembang, and Grobogan Regencies. Increasing population pressure and limited diversification of the agricultural economy are the main factors maintaining high poverty rates in these regions. The consistent high poverty clusters in the northeast indicate the existence of a structural poverty trap, where economic development is uneven and regional integration with centers of economic growth remains low. The resulting pattern demonstrates that poverty in Central Java is spatial, with poor areas tending to be close to other poor areas due to similar socio-economic conditions and limited infrastructure across regions.

Fig. 5 depicts the spatial distribution of the Food Security Index (FSI) for the 2023–2024 period. In 2023, regions with high FSI scores or good food security were concentrated in the northern and central corridors, such as Semarang, Kendal, Demak, and Kudus. These areas benefit from their proximity to major trade routes (the Pantura route), the availability of transportation and markets, and irrigation networks that support agricultural productivity. In contrast, the southern and mountainous regions (Banjarnegara, Wonosobo, and Kebumen) have low FSI scores, indicating a persistently high level of food vulnerability due to limited productive land, difficult distribution access, and low local food diversification

In 2024, the FSI map (bottom right) shows an increase in food security scores in several central and eastern regions, such as Grobogan, Pati, and Blora, indicating the effectiveness of area-based policy interventions. Programs such as village food barns, strengthening staple food distribution, and developing local food commodities appear to be beginning to have a positive impact on food availability and price stability. However, the southern mountainous region still shows low FSI scores, indicating that food vulnerability in this region is geographically and recurrently affected by biophysical conditions, accessibility, and limited economic infrastructure.

Maps of the distribution of poverty and food security generally show a negative spatial relationship between the two variables: areas with high poverty rates generally overlap with areas with low food security, and vice versa. These findings are consistent with the results of Moran's I

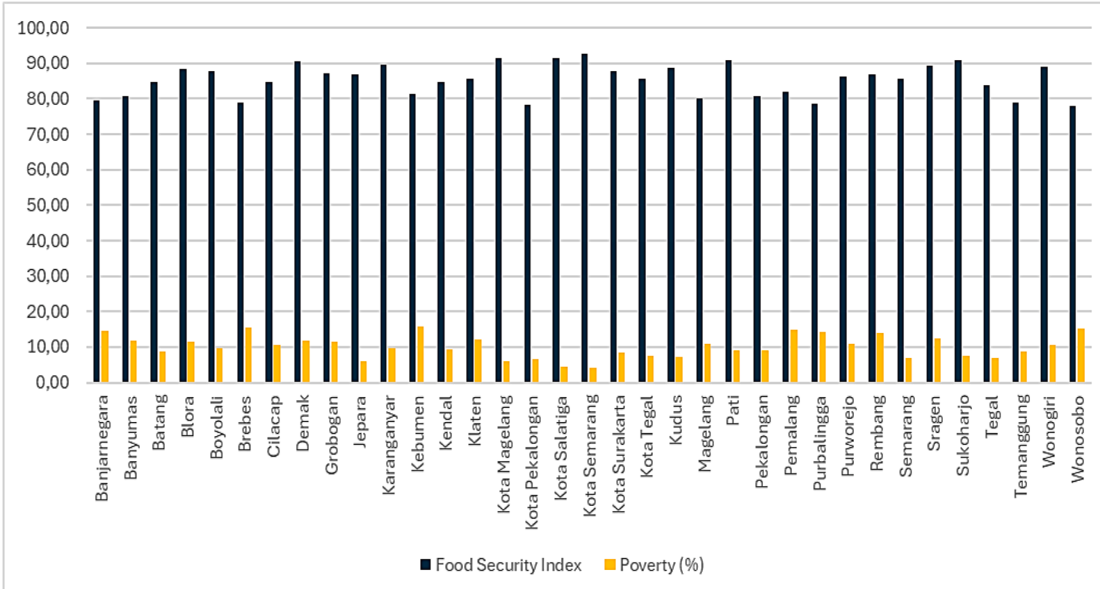

Fig. 3. Percentage of poverty and food security index of central java province in 2024.
source: central statistics agency and national food agency, 2025.

and LISA analyses, which showed significant spatial autocorrelation across several regional clusters. Thus, poverty and food security in Central Java are determined not only by economic and production factors, but also by spatial connectivity between regions. Therefore, improving food security and poverty alleviation needs to be designed within a place-based policy framework, which takes into account interregional connectivity, geographic conditions, and local economic capacity in an integrated manner.

4.2. Spatial pattern of distribution of poverty and food security index in central java province

A spatial analysis of poverty and food security in Central Java in 2023–2024 was conducted to understand how distribution patterns between regions are interconnected. Using Moran's Index (univariate and bivariate) and the Local Indicator of Spatial Association (LISA), this study was able to identify whether the distribution of poverty and food security is clustered, dispersed, or random. Furthermore, bivariate analysis was used to determine the spatial relationship between poverty and the food security index, thus obtaining an overview of which regions fall into the High-High, Low-Low, High-Low, or Low-High categories. The results of the analysis were then visualized in the form of scatterplots and spatial maps to provide a more comprehensive understanding of the phenomena occurring in Central Java (Table 3).

The results of the univariate analysis of Moran's I indicate that both poverty and food security variables in Central Java Province still exhibit clustered spatial patterns, indicating a geographic interconnectedness between regions in terms of community welfare. This pattern indicates that the socio-economic conditions of a district do not stand alone, but are influenced by the conditions of the surrounding areas. In 2023, the Moran's I value of 0.288 ($p < 0.01$) for the poverty variable indicates a strong positive spatial autocorrelation, where areas with high poverty rates tend to be close to other areas with high poverty rates. This phenomenon illustrates the spatial convergence of poverty, which can be explained by the theory of *spatial dependence,* namely when economic limitations, infrastructure, and access to public resources in one region also affect neighboring regions through socio-economic interactions and local trade networks. However, in 2024, the autocorrelation value decreased to 0.184 ($p < 0.05$), indicating that the distribution of poverty is becoming more widespread and less clustered. This change can be attributed to relatively equitable development dynamics and increased economic mobility between regions. Regional poverty alleviation programs such as integrated social assistance, MSME empowerment, and increased interregional connectivity in Central Java have the potential to narrow this spatial economic gap. In other words, there has been a shift from a concentrated poverty pattern to a more dispersed one, indicating a process of welfare decentralization in several regions.

The food security index (FSI) variable shows a relatively consistent but weaker pattern than poverty, with a Moran's I value of 0.164 ($p < 0.05$) in 2023 and decreasing to 0.113 (not significant) in 2024. These results indicate that food security has a more complex spatial dimension, determined not only by economic factors but also by environmental factors, agricultural land availability, access to distribution infrastructure, and regional food policies. The cluster pattern in food security in 2023 reflects that areas with low food security conditions are generally adjacent to poor areas, especially in the southern and interior regions of Central Java which still rely on the subsistence agricultural sector and have limited logistical infrastructure.

The decline in autocorrelation strength in 2024 can be interpreted as the beginning of changes in the spatial structure of food security in Central Java Province. This condition indicates that several regions are beginning to show increased food self-sufficiency through economic diversification, strengthening food logistics systems, and implementing area-based food security programs promoted by the provincial government. For example, through optimizing *village food barns,* strengthening Village-Owned Enterprises (BUMDes), and increasing the efficiency of agricultural product distribution across districts. These efforts have encouraged several previously food-insecure regions to become more self-sufficient in the provision and distribution of staple foods.

However, the decline in spatial cluster strength also reflects the increasingly pronounced heterogeneity of food security across regions. The most significant differences are observed between coastal and mountainous areas, two key landscapes in Central Java with very different ecological and economic characteristics. Northern coastal areas, such as Demak, Pekalongan, and Pati Regencies, have relatively better access to trade networks, transportation infrastructure, and food markets. Geographical proximity to the northern coastline (Pantura) and national food distribution centers allows coastal areas to adapt more quickly to food supply and price dynamics. On the other hand, coastal areas also face unique risks, such as vulnerability to tidal flooding,

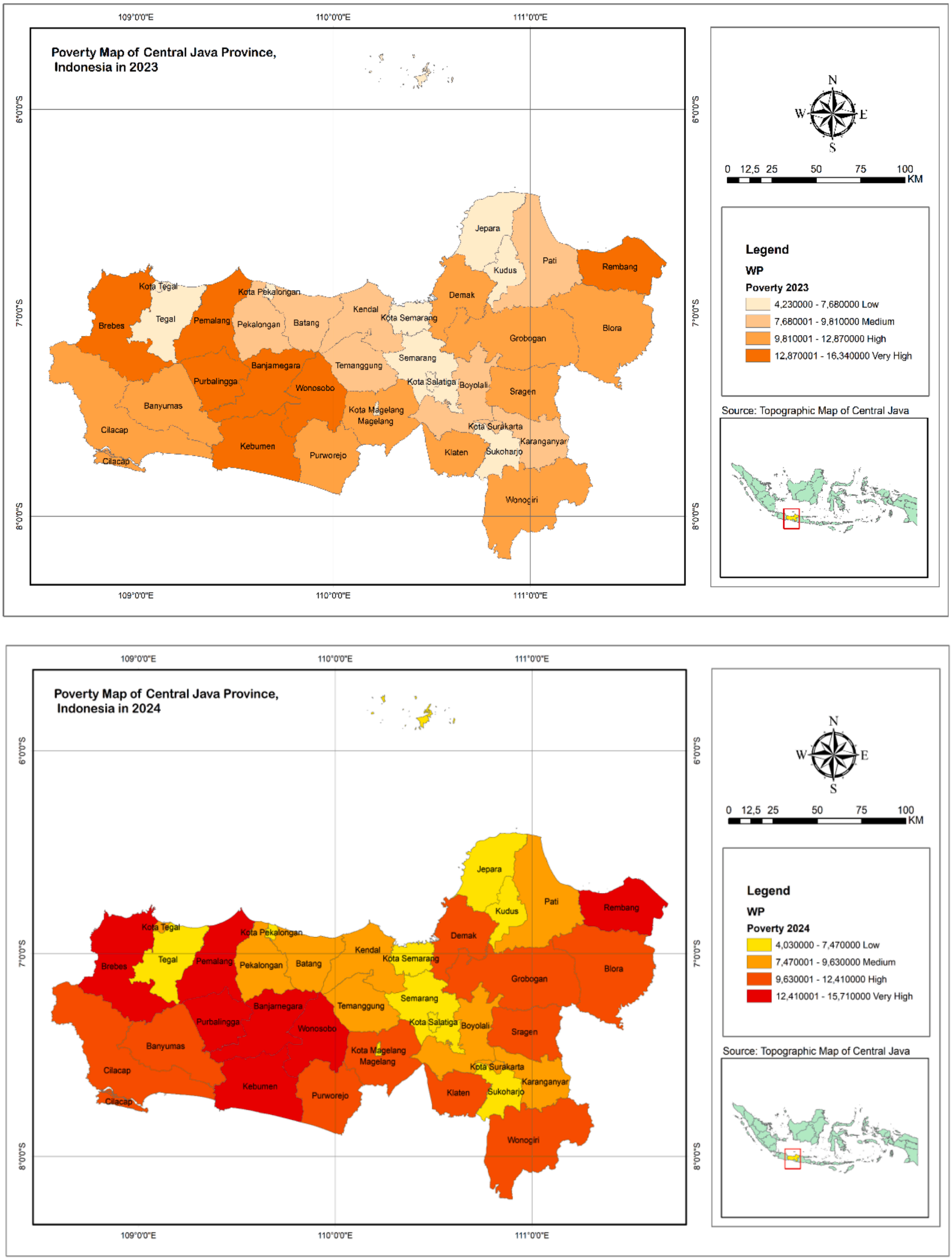

Fig. 4. Map of the distribution of the poverty in central java province in 2023–2024.
source: Central statistics agency, 2025.

seawater intrusion, and pond degradation, which can disrupt local agricultural production and threaten food security if not supported by a robust logistics system.

In contrast, mountainous and inland areas such as Banjarnegara, Wonosobo, and Kebumen face different structural challenges. Although these areas possess relatively fertile natural resources suitable for horticultural agriculture, steep topography and limited transportation infrastructure lead to high distribution costs and dependence on

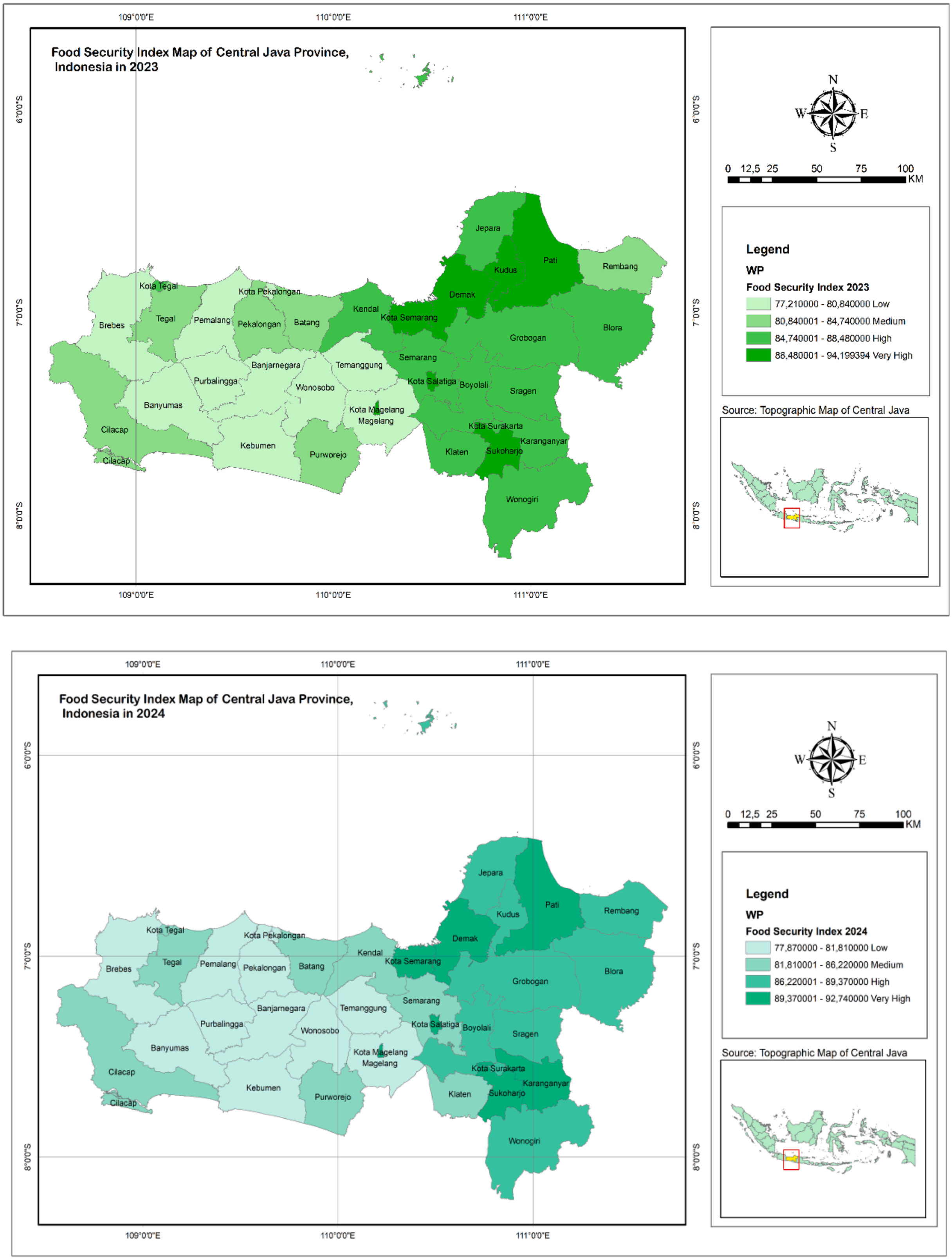

Fig. 5. Map of the distribution of the food security index in central java province in 2023–2024.
Source: National food agency, 2025.

interregional markets. Furthermore, land productivity in mountainous areas is highly dependent on seasons and water availability, making communities more vulnerable to weather fluctuations and climate change. Spatial-wise, this explains why mountainous regions often form low-low food security clusters in LISA analyses, while coastal and urban areas tend to have high-high food security clusters.

These ecological and economic differences between coastal and mountainous areas demonstrate that the disparity in food security in

Table 3
Results of Moran's I univariate analysis for poverty and food security variables in central java (2023–2024).

Variables	Year	Moran's I value	z-value	p-value	Interpretation
Poverty Percentage	2023	0.288	2.8269	0.0090	There is significant positive spatial autocorrelation; areas with high poverty rates tend to be close to other poor areas.
Poverty Percentage	2024	0.184	1.8340	0.0390	The positive spatial autocorrelation is weak but significant; poverty patterns still show interregional linkages but are beginning to weaken.
Food Security Index	2023	0.164	1.7267	0.0470	There is significant positive spatial autocorrelation; areas with high food security tend to be close together, as do areas with food insecurity.
Food Security Index	2024	0.113	1.2687	0.1030	Not statistically significant; the distribution of food security is becoming more random and does not show strong spatial clustering.

Source: Spatial Analysis Output, 2025.

Central Java is not solely due to economic factors, but also to biophysical and spatial conditions that influence food access and availability. Therefore, policies to improve food security cannot be designed with a *one-size-fits-all approach* but must be adaptive to local geographic and socioeconomic characteristics. In coastal areas, strengthening port infrastructure, controlling abrasion, and integrating the food supply chain are priorities. Meanwhile, in mountainous areas, policies that emphasize land conservation, sustainable irrigation, and strengthening local agricultural community-based economies are more relevant.

4.3. Results of the bivariate LISA analysis between poverty and the food security index in central java (2023–2024)

The LISA Bivariate Cluster Map analysis shows that in 2023, the spatial patterns between poverty and the food security index (FSI) in Central Java Province formed diverse cluster configurations, reflecting the complex spatial relationship between economic well-being and food security conditions across regions. The High–High cluster, found in Pati and Kudus Regencies, depicts areas with high poverty rates surrounded by neighbors with high food security. This condition indicates internal disparities, where poor areas are located around areas with relatively good food security. This may reflect inequalities in food access and distribution, despite adequate regional food availability. This phenomenon also indicates that the poverty problem in the region is structural, not solely due to a lack of food supply, but related to limited purchasing power and economic inclusion of the poor within a stronger regional food system (Table 4).

In contrast, the High–Low cluster, encompassing Brebes, Tegal, Pemalang, Batang, and Jepara, depicts areas with high poverty surrounded by neighbors with low food security. This pattern indicates a condition of dual spatial vulnerability, where high poverty and low food security tend to co-occur and reinforce spatial disadvantage. These areas are generally located on the western and northern coasts of Central Java, characterized by small-scale agriculture, high dependence on weather, and limited logistical infrastructure. This *High–Low phenomenon* underscores the importance of cross-regional interventions, as economic downturns in one region have the potential to exacerbate food insecurity in neighboring areas.

The Low–High cluster, found in Grobogan and Blora Regencies, indicates areas with low poverty surrounded by neighbors with high food security. This condition indicates a spatial advantage, where regions with better economic conditions receive support from a network of neighboring regions with strong food systems. In other words, this cluster has the potential to become a buffer zone that can support regional food stability due to its greater adaptive capacity and economic capacity.

Interestingly, no Low-Low clusters were found, meaning areas with low poverty adjacent to neighbors with low food security. The absence of this pattern indicates that prosperous areas in Central Java are generally adjacent to areas with relatively good food security, meaning positive *spatial diffusion* is more dominant in areas with high economic power. Overall, in 2023, the cluster pattern shows a heterogeneous spatial trend, with a combination of poor areas trapped around high-resilience areas and poor areas adjacent to each other in vulnerable coastal areas.

In 2024, the results show a significant decrease in the number of clusters, indicating a weakening of spatial autocorrelation between variables. The High-High cluster persisted only in Pati Regency, indicating the persistence of structural poverty amidst relatively strong food

Table 4
Results of the LISA bivariate cluster map analysis between poverty and the food security index in central java (2023–2024).

Year	Cluster Category	Areas Included	Interpretation
2023	High-High (2)	Pati Regency, Kudus Regency	Areas with high poverty rates adjacent to areas with high food security show a strong positive correlation between well-being and food availability.
2023	High–Low (5)	Brebes, Tegal, Pemalang, Batang, Jepara Regencies	Areas with high poverty but surrounded by areas with low food security indicate the potential for cross-regional economic spillover effects.
2023	Low–High (2)	Grobogan Regency, Blora	Areas with low poverty adjacent to areas with high food security
2023	Low–Low (0)	–	No region had a consistently low pattern on both variables.
2023	Not Significant (26)	26 other districts/cities (including Semarang, Banyumas, Banjarnegara, Magelang, Wonosobo, Sragen, and others)	Shows no significant spatial pattern; poverty and food security tend to be randomly distributed without strong autocorrelation.
2024	High-High (1)	Pati Regency	The region remains in a state of high vulnerability across two variables, indicating the persistence of structural problems of poverty and food security.
2024	High–Low (3)	Brebes, Tegal, Pemalang Regencies	Poor areas and surrounded by areas with low food security
2024	Low–High (2)	Jepara Regency, Batang	Areas with low poverty rates but adjacent to areas with high food security need to anticipate external impacts on local food systems.
2024	Low–Low (0)	–	No stable low pattern was found in the two variables.
2024	Not Significant (27)	27 other districts/cities (including Grobogan, Kudus, Blora, Magelang, Banyumas, and others)	Spatial pattern is not significant; shows random distribution between regions.

Source: Spatial Analysis Output, 2025.

security in neighboring regions. This suggests that poverty in Pati is not primarily associated with food supply deficits, but rather coexists with economic and social constraints, but rather by economic and social barriers such as limited access to formal employment and production infrastructure. Meanwhile, the High-Low cluster persisted in several west coast areas such as Brebes, Tegal, and Pemalang, indicating areas experiencing dual pressures due to weak local economic systems and environmental disturbances such as tidal flooding and agricultural land degradation.

On the other hand, the Low–High cluster is still visible in Jepara and Batang Regencies, indicating regions with better economic well-being but also benefiting from being located near areas with high food security. This condition reinforces the role of these regions as centers of regional food stability amidst economic and climate fluctuations. Meanwhile, the Low–Low cluster was not found in 2024, reinforcing the finding that low poverty patterns are generally associated with strong surrounding food systems.

The pattern revealed by the LISA Bivariate analysis generally from 2023 to 2024 shows a spatial shift from a collective vulnerability pattern (High–Low) to a more dispersed and fragmented pattern, reflecting the complex socio-economic dynamics in Central Java. This shift can be attributed to the enhancement of regional food security programs, strengthening interregional connectivity, and the growth of the non-agricultural sector in coastal and central areas. However, the persistence of the High–High cluster in Pati and the High–Low cluster on the west coast indicates that food security policies need to be more directed at strengthening the economic base of poor households, rather than simply increasing food supplies at the regional level. (Table 5)

Table 5
Results of the bivariate LISA significance map analysis between poverty and the food security index in central java (2023–2024).

Year	Significance Category	Areas Included	Interpretation
2023	$p = 0.05$ (3 areas)	Grobogan, Blora, Kudus Regencies	Areas with a weak spatial relationship between poverty and food security are beginning to show a link, but it is not yet widely significant.
2023	$p = 0.01$ (6 regions)	The Regencies of Pati, Brebes, Tegal, Pemalang, Batang, Jepara	The spatial relationships are significant; these areas are the main clusters of interactions between poverty and food vulnerability.
2023	$p = 0.001$ (0 areas)	–	There are no areas with very strong spatial relationships.
2023	Not Significant (26 regions)	Other regencies/cities in Central Java (including Semarang, Sragen, Banjarnegara, Wonosobo, and others)	Does not show significant spatial correlation between poverty and food security.
2024	$p = 0.05$ (5 areas)	Brebes, Tegal, Pemalang, Pati, Jepara Regencies	Areas with weak spatial relationships are significant; some are a continuation of the previous year's pattern.
2024	$p = 0.01$ (3 areas)	Batang, Pekalongan, Grobogan Regencies	The spatial relationship is quite strong; it reflects a shift in spatial focus from east to west of Central Java.
2024	$p = 0.001$ (0 areas)	–	There are no areas with very strong spatial relationships.
2024	Not Significant (27 regions)	27 other districts/cities (including Blora, Banyumas, Wonosobo, and Sragen)	Does not show a meaningful spatial relationship between the two variables.

Source: Spatial Analysis Output, 2025.

The Bivariate LISA Significance Map results reinforce the findings of the cluster map by showing varying levels of spatial significance across regions and periods. In 2023, most regencies in Central Java (26 regions) were categorized as not significant, indicating that the spatial relationship between poverty and food security in most regions was still weak or random. However, there were 6 regions with a significance level of $p = 0.01$, namely Pati, Brebes, Tegal, Pemalang, Batang, and Jepara, which showed a fairly strong spatial relationship between high poverty and low food security. These regions are generally located in the northern coastal region with agricultural characteristics, high population density, and dependence on agricultural systems that are vulnerable to climate disturbances. Meanwhile, Grobogan, Blora, and Kudus were included in the $p = 0.05$ group, indicating a weak spatial relationship but beginning to show a consistent pattern between social conditions and food. No region reached a significance level of $p = 0.001$, indicating that no extreme spatial correlation was found, so the relationship between regions remains moderate.

In 2024, the results show a decrease in the number of significant regions and a shift in the pattern toward the central and western regions of Central Java. Regions with a significance level of $p = 0.05$ include Brebes, Tegal, Pemalang, Pati, and Jepara, while $p = 0.01$ is found in Batang, Pekalongan, and Grobogan. This shift indicates that the spatial dynamics between poverty and food security are beginning to shift geographically, likely influenced by the development of trade infrastructure, improvements in regional connectivity, and the implementation of local economic empowerment programs. Furthermore, the insignificant increase in regions in 2024 indicates a decrease in the spatial concentration of extreme poverty and an increase in inter-regional variation, indicating that some regions have succeeded in strengthening food security, although some regions remain lagging behind. (Fig. 6)

The LISA Significance Map results generally indicate that the spatial relationship between poverty and food security in Central Java is dynamic, with the strength of the relationship changing with policy interventions and changes in local socioeconomic conditions. This finding underscores the importance of a spatial approach in formulating regional development policies, as the effects of poverty and food security are not limited to within a district but can also spread externally to surrounding areas.

4.4. Discussion dan research implications

The results of this study demonstrate that poverty and food security in Central Java have a significant spatial relationship. Univariate Moran's I analysis in 2023 and 2024 revealed a positive spatial autocorrelation pattern in both the distribution of poverty and food security. This means that areas with high poverty rates tend to cluster with other poor areas, while areas with low food security also cluster with similar areas. This finding is consistent with the basic theory of spatial dependence, which emphasizes that the socio-economic conditions of a region do not stand alone but are influenced by its proximity to other regions.

The results of the Bivariate Moran's I test indicate a negative spatial correlation between poverty levels and food security. Areas with high poverty rates tend to be associated with low food security indices, and vice versa. This pattern is also clearly visible in the Bivariate LISA map, which classifies Central Java into High-High (high poverty–high food security), Low-Low (low poverty–low food security), and a combination of High-Low and Low-High. The significance map confirms that this relationship is not coincidental, but rather a real, statistically significant pattern.

The results of this study align with those of Hyman et al. [21], who emphasized the importance of spatially mapping poverty and food security to support area-based policies. They asserted that a spatial approach can reveal unequal distribution of resources and food security that cannot be captured by conventional analysis. Ara & Ostendorf's

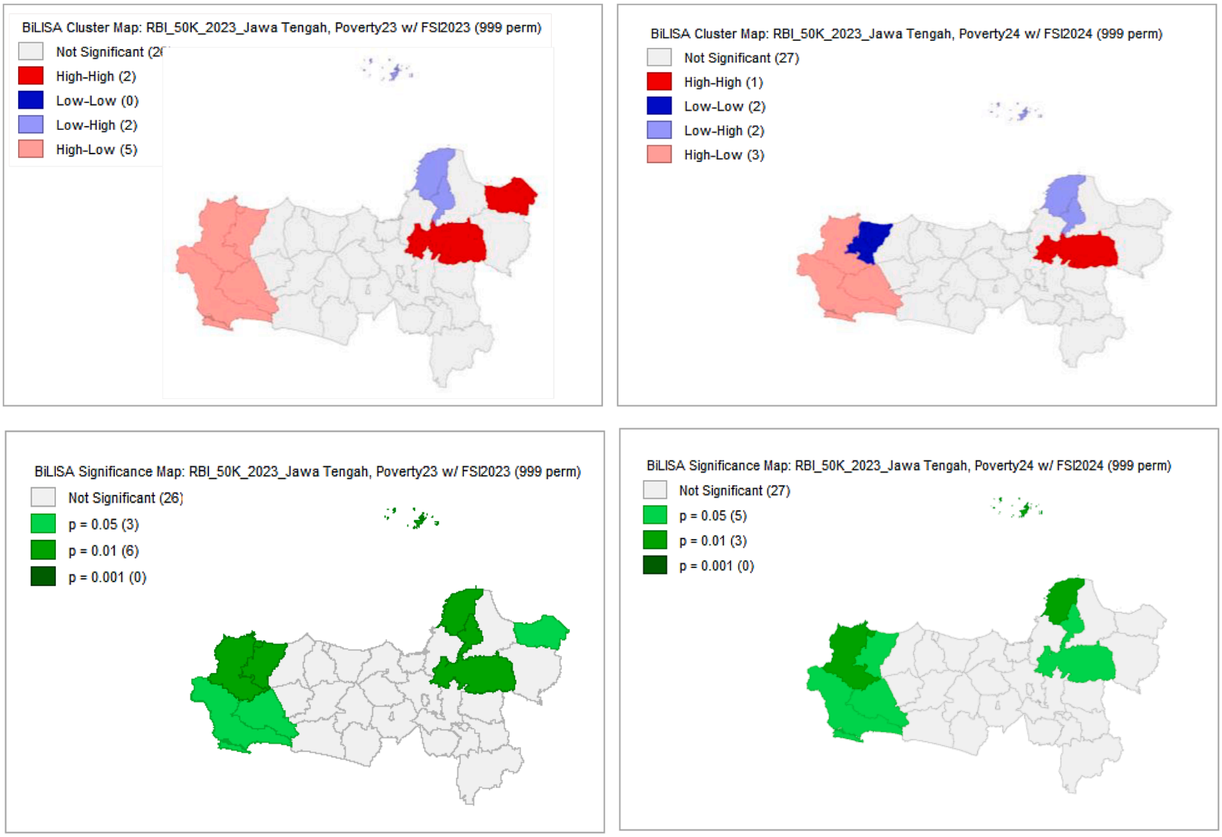

Fig. 6. LISA results map bivariate cluster map and bivariate LISA significance map of poverty and food security index of central java province in 2023–2024. source: Spatial analysis output, 2025.

[22] study in Bangladesh also demonstrated the urgency of developing spatially based food policies. Their findings are similar to those in Central Java, where poverty is strongly associated with low food security. This underscores the need for a cross-regional approach in formulating policy interventions, rather than just a sectoral approach. Furthermore, Odhiambo et al.'s [23] study in Kenya on food diversification confirmed that household food security is strongly influenced by geographic conditions and the spatial distribution of agricultural land. These findings are relevant to the pattern found in Central Java, where rural areas with high poverty and limited access to food are grouped as High-High on the LISA map.

Brown [24] and Virtriana et al. [25] emphasize that the use of spatial technologies such as remote sensing and high-resolution spatial data can identify food-insecure areas more precisely. In this study, the use of Moran's I and LISA demonstrated the effectiveness of spatial methods in uncovering patterns of links between poverty and food security previously hidden in aggregate data.

Vonthron et al.'s [26] study on the foodscape concept is also relevant to understanding the patterns found. They argue that food security is not only related to food availability but also to spatial contexts such as access, distribution, and socioeconomic networks. Mapping the High-High cluster in Central Java shows that food-insecure areas are influenced not only by production factors but also by accessibility and community welfare. Research by Vasa et al. [27] comparing spatial aspects of food security in Azerbaijan, Hungary, Austria, Singapore, and Georgia found that geographic factors, infrastructure, and economic distribution significantly influence regional food security. This aligns with conditions in Central Java, where differences between regions, both in terms of geography (coastal, inland, urban) and economic infrastructure, create disparities in achieving food security.

However, unlike previous studies that reported strong positive spatial clustering, the bivariate Moran's I statistic in this study indicates a negative spatial correlation between poverty and food security across districts in Central Java. This result demonstrates a diffuse spatial pattern, suggesting that areas with high poverty rates are not necessarily geographically clustered with low levels of food security. This divergence underscores the context-specific nature of spatial dynamics in Central Java, where socioeconomic and geographic disparities particularly between coastal and inland areas play a decisive role. Coastal districts, despite exhibiting relatively higher poverty rates, tend to maintain stronger food security due to diversified economic activities and better market access. Conversely, some inland agricultural areas experience food insecurity driven by limited infrastructure, market isolation, and vulnerability to climate variation. This spatial dispersion highlights that the relationship between poverty and food security in Indonesia is not uniform, but rather spatially heterogeneous and asymmetrical. Therefore, the contribution of this study lies in providing district-level empirical evidence that challenges the assumption of positive spatial clustering observed in other developing countries. The results of this study emphasize the need for a differentiated and decentralized policy strategy across regions, where poverty alleviation and food security programs are tailored to the socioeconomic and geographic characteristics of each region. Such an approach is crucial for achieving balanced regional development and enhancing local resilience within Indonesia's decentralized governance framework.

The results of this study reinforce the theory of socio-spatial inequality [28], which states that food security has both social and spatial dimensions. The concentration of poor areas with low food security indices demonstrates spatial inequality that requires a focused intervention approach. These findings also emphasize the importance of utilizing digital technology and big data to support spatial planning in the Agriculture 4.0 era [29]. Integrating spatial analysis with machine learning or predictive models can help local governments anticipate future food security.

Furthermore, these results confirm the findings of Amirzadeh Moradabadi et al. [30], which showed that agricultural sustainability in Iran is directly related to household food security. In the context of Central Java, the sustainability of rural food production is also significantly influenced by poverty levels. Therefore, increases in agricultural productivity are not necessarily accompanied by improvements in food security when poverty levels remain high.

The findings of this study support the importance of spatial intervention strategies to reduce inter-regional disparities. Local governments can use the LISA analysis results as a basis for determining intervention priorities, for example by paying special attention to areas categorized as High-High. As noted by Grace et al. [31], a spatial analysis framework is very helpful in designing more efficient and targeted food aid distribution. Furthermore, this study's findings emphasize the urgency of more adaptive food policies at the local level. Thow et al. [32] noted that Indonesia's food policy faces limited latitude due to global trade regulations, making a spatially based strategy at the provincial level a realistic alternative solution.

The findings of this study have important implications for the formulation of food security policies in Central Java Province. The spatial analysis reveals a disparity between coastal and inland areas, requiring different policy approaches based on their geographic and socioeconomic characteristics. In northern coastal areas such as Brebes, Pekalongan, and Demak, food security tends to be more influenced by economic fluctuations and limited logistics infrastructure, particularly in the supply chain and storage of agricultural products. Therefore, policies need to be directed at strengthening food distribution systems, investing in cold chain infrastructure, and modernizing traditional markets to maintain stable food supplies and prices. Conversely, in inland and mountainous areas such as Wonosobo, Banjarnegara, and Purworejo, the main challenges lie in limited physical access, transportation infrastructure, and low diversification of household food sources. Therefore, community empowerment programs based on local food, farmer capacity building through sustainable agricultural training, and the development of production roads and village irrigation are needed. The integration of these two approaches can be supported through the implementation of the Regional Action Plan for Food and Nutrition (RAD-PG) and synergy with the Central Java Regional Food Security program 2025–2030, so that the resulting policies not only reduce poverty levels, but also strengthen food security in a sustainable and equitable manner throughout the province.

5. Conclusion

The findings of this study reveal that poverty and food security in Central Java Province during the 2023–2024 period exhibit a significant yet spatially uneven relationship. The results of the univariate Moran's I analysis confirm that both poverty and food security are not randomly distributed but rather display clustered spatial patterns, indicating persistent spatial inequality in community welfare. The bivariate Moran's I analysis further identifies a negative spatial correlation, suggesting that areas with high poverty levels are often adjacent to regions with relatively higher food security, reflecting a dispersed pattern rather than uniform clustering. These findings reflect spatial association rather than causal relationships. This nuanced result underscores the complex and heterogeneous spatial dynamics shaping poverty–food security interactions at the regional level.

Beyond its empirical findings, this study makes a methodological contribution by advancing the application of spatial analytical tools specifically Moran's I and Local Indicators of Spatial Association (LISA) to examine socio-economic linkages across subnational contexts. By employing these techniques at the district level, the research demonstrates how spatial econometric methods can capture local disparities that traditional non-spatial models might overlook. Theoretically, the results contribute to a deeper understanding of territorial asymmetry in the poverty and food security nexus within decentralized governance frameworks.

From a policy perspective, the study highlights the need for spatially differentiated and region-based strategies in Central Java. Policymakers should prioritize coastal districts facing logistical and infrastructure bottlenecks while simultaneously strengthening food diversification and agricultural resilience in inland areas. Integrating spatial evidence into provincial development planning will enhance the effectiveness of poverty reduction and food security programs, ensuring that interventions are geographically targeted, data-driven, and contextually adaptive to the unique socio-economic landscape of Central Java.

Based on these findings, this study recommends several things. First, local governments need to focus on High-High areas with integrated intervention programs that combine poverty alleviation and strengthening food access. Second, strategies to improve food security should not only focus on production aspects, but also pay attention to distribution, accessibility, and affordability of food prices in vulnerable areas. Third, the integration of spatial and digital technologies such as predictive modeling, remote sensing, and big data analytics is needed to monitor the dynamics of food security and poverty in real time. Furthermore, further research is recommended to examine in more depth other factors that influence the link between poverty and food security, such as education, infrastructure, and climate change. This study has methodological limitations that must be acknowledged, particularly related to the scope of the variables analyzed. The study focused on only one independent variable, namely the poverty rate, and one dependent variable, namely the Food Security Index (FSI). This approach facilitates spatial analysis and allows for the exploration of patterns of direct relationships between poverty and food security at the district/city level. Therefore, the results obtained should be interpreted cautiously, considering the limited scope of the variables. Further research is recommended to integrate these structural and environmental variables into a spatial analysis framework to provide a more comprehensive understanding of the determinants of food security at the regional level.

CRediT authorship contribution statement

Shanty Oktavilia: Writing – review & editing, Writing – original draft. **Firmansyah:** Writing – review & editing, Writing – original draft. **Zainur Hidayah:** Writing – review & editing, Writing – original draft. **Fauzul Adzim:** Writing – review & editing, Writing – original draft. **Sucihatiningsih Dian Wisika Prajanti:** Writing – review & editing, Writing – original draft. **Fafurida:** Writing – review & editing, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Shanty Oktavilia reports financial support was provided by Universitas Negeri Semarang. Shanty Oktavilia reports a relationship with Universitas Negeri Semarang that includes: employment and funding grants. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors express their deepest appreciation and gratitude to the Indonesian Collaborative Research Organizing Institute (RKI) in 2025 for its financial support for this research. The assistance and facilitation provided through the RKI program were instrumental in ensuring the smooth running of data collection, analysis, and the preparation of research results. This support not only contributed to the completion of this research but also strengthened research collaboration between universities in Indonesia in the pursuit of science-based sustainable development.

References

[1] M. Mathenge, B.G.J.S. Sonneveld, J.E.W. Broerse, Mapping the spatial dimension of food insecurity using GIS-based indicators: a case of Western Kenya, Food Secur. 15 (2023) 243–260, https://doi.org/10.1007/s12571-022-01308-6.
[2] M.R. Pakravan-Charvadeh, H.A. Khan, C. Flora, Spatial analysis of food security in Iran: associated factors and governmental support policies, J. Public Health Policy 41 (2020) 351–374, https://doi.org/10.1057/s41271-020-00221-6.
[3] A.C.T. Rosalia, S, L. Hakim, Spatial analysis of the impact of flood and drought on food Security index, Nat. Environ. Pollut. Technol. 20 (2021), https://doi.org/10.46488/NEPT.2021.v20i02.031.
[4] P. Ataei, H. Sadighi, N. Izadi, Major challenges to achieving food security in rural Iran, Rural Soc. 30 (1) (2021) 15–31.
[5] M. Hlatshwayo, T.P. Mbombo-Dweba, Food security status of Zimbabwean immigrants living in Msunduzi municipality, South Africa during the COVID-19 pandemic, Asian Dev. Policy Rev. 12 (4) (2024) 378–395.
[6] D. Tleshpayeva, N. Bondarenko, M. Leontev, G. Mashentseva, J. Plaksa, A. Zharov, A. Karbozova, Assessment of economic management of land resources to enhance food security, Qubah. Acad. J. 5 (1) (2025) 159–168.
[7] S.A. Lone, I.A. Mayer, Geo-spatial analysis of land use/land cover change and its impact on the food security in District Anantnag of Kashmir Valley, GeoJournal. 84 (2019) 785–794, https://doi.org/10.1007/s10708-018-9891-2.
[8] Xuanzhu Meng, Linqi. Liu, Spatial-temporal evolution and influencing factors of food production security in China's main food-producing areas– based on resource constraint perspective, Ecol. Indic. 171 (2025) 113224.
[9] Z.A. Alemu, A.A. Ahmed, A.W. Yalew, B. Simanie, Spatial variations of household food insecurity in East Gojjam Zone, Amhara Region, Ethiopia: implications for agroecosystem-based interventions, Agric. Food Secu. 6 (2017), https://doi.org/10.1186/s40066-017-0113-9.
[10] J. Cai, E. Ma, J. Lin, L. Liao, Y. Han, Exploring global food security pattern from the perspective of spatio-temporal evolution, J. Geogr. Sci. 30 (2020) 179–196, https://doi.org/10.1007/s11442-020-1722-y.
[11] A. Bofa, T. Zewotir, A bayesian spatio-temporal dynamic analysis of food security in Africa, Sci. Rep. 14 (2024) 15132, https://doi.org/10.1038/s41598-024-65989-z.
[12] Wilson Nguru, et al., Spatial estimation of flood residual water cultivation (FRWC) potential for food security in Sédhiou and Tambacounda regions of Sénégal, Agric. Water. Manage 287 (2023) 108445.
[13] B. Feizizadeh, S. Pourmoradian, S. Pourmoradian, Food security assessment based on GIS spatial analysis in the rural area of East Azerbaijan Province, Iran, GI_Forum 1 (2015), https://doi.org/10.1553/giscience2015s475.
[14] A. Hameed, S. Karim, I.U.H. Padda, Spatial analysis of food and nutrition security in Pakistan: a holistic pathway towards zero hunger policies, GeoJ. 88 (2022), https://doi.org/10.1007/s10708-022-10762-w.
[15] J. Qiao, Q. Cao, Z. Zhang, Z. Cao, H. Liu, Spatiotemporal changes in the state of food security across mainland China during 1990–2015: a multi-scale analysis, Food Energy Secu. 11 (2022), https://doi.org/10.1002/fes3.318.
[16] F. Lv, Z. Wang, Q. Wu, J. Qiao, Z. Zhang, L. Deng, Multiscale analysis of factors affecting food security in China, 1980-2017, Environ. Sci. Pollut. Res. 29 (2021), https://doi.org/10.1007/s11356-021-16125-1.
[17] E.C. Merem, A. Hines, Y. Twumasi, D. Foster, C. Romorno, D. Olagbegi, et al., Regional assessment of the food Security situation in West Africa with GIS, Food Public Heal 9 (2019), https://doi.org/10.5923/j.fph.20190902.04.
[18] K. Yadav, R.G. Congalton, Accuracy assessment of global food security-support analysis data (GFSAD) cropland extent maps produced at three different spatial resolutions, Remote Sens. 10 (2018), https://doi.org/10.3390/rs10111800.
[19] I.Y. Safitri, Ruliana-, M.A. Tiro, Spatial regression analysis to see factors affecting food security at district level in South Sulawesi Province, ARRUS J. Math. Appl. Sci. 2 (2022), https://doi.org/10.35877/mathscience740.
[20] G. Hyman, C. Larrea, A. Farrow, Methods, results and policy implications of poverty and food security mapping assessments, Food Policy. 30 (2005), https://doi.org/10.1016/j.foodpol.2005.10.003.
[21] S. Amalia, S. Maria, D. Fitrianitha, The dynamics of food security in Indonesia: a study on the availability, affordability, and utilization of food in districts and cities with the lowest index in 2022, Int. J. Econ. (IJEC) 3 (2024), https://doi.org/10.55299/ijec.v3i2.1022.
[22] E. Meiyetriani, B. Utomo, D.N. Iswarawanti, Socio-economic inequality factors predicting food Security index in 514 districts of Indonesia: a supervised machine learning regression analysis, Malays. J. Med. Health Sci. 21 (2025).
[23] C.M. Firdausy, D.A. Budisetyowati, Variables, dimensions, and indicators important to develop the multidimensional poverty line measurement in Indonesia, Soc. Indic. Res. 2 (2022), https://doi.org/10.1007/s11205-021-02859-5.
[24] P.A. Moran, Notes on continuous stochastic phenomena, Biometrika 37 (1950).
[25] Ghosh, P., Saha, D., & Cartone, A. Housing inequality by social groups, gender, and place of residence: a district-level geospatial analysis in India. In Accessible Housing For South Asia: Needs, Implementation and Impacts. 2022. https://doi.org/10.1007/978-3-030-88881-7_4.
[26] I. Ara, B. Ostendorf, A review of food security and the potentials to develop spatially informed food policies in Bangladesh, Earth. Syst. Environ. 1 (2017), https://doi.org/10.1007/s41748-017-0021-y.
[27] O. Odhiambo, M. Maingi, N. Njeri, Spatial analysis of food crop diversification in Busia County-Kenya: implications on household food security, World J. Agric. Sci. Technol. 2 (2024), https://doi.org/10.11648/j.wjast.20240202.13.
[28] M.E. Brown, Remote sensing technology and land use analysis in food security assessment, J. Land. Use Sci. 11 (2016), https://doi.org/10.1080/1747423x.2016.1195455.
[29] R. Virtriana, T.S. Anggraini, K. Wikantika, K.T.N. Ihsan, F.C. Mustika, B. Soeksmantono, et al., Development of spatial model for food security prediction using remote sensing data in West Java, Indonesia. ISPRS Int. J. Geo-Inf. 11 (2022), https://doi.org/10.3390/ijgi11050284.
[30] S. Vonthron, C. Perrin, C.-T. Soulard, Foodscape: a scoping review and a research agenda for food security-related studies, PLoS. One 15 (2020), https://doi.org/10.1371/journal.pone.0233218.
[31] L. Vasa, R. Huseynov, I. Varga, L. Dávid, The regional and geographical aspects of food security: a spatial analysis In The case of Azerbaijan, Hungary, Austria, Singapore and Georgia, Geogr. Tech. 15 (2020), https://doi.org/10.21163/gt_2020.152.16.
[32] T. Chagomoka, A. Drescher, S. Unger, R. Glaser, J. Schlesinger, B. Marschner, Food coping strategies in northern Ghana. A socio-spatial analysis along the urban–rural continuum, Agric. Food Secu. 5 (2016), https://doi.org/10.1186/s40066-016-0052-x.